

GuardianIQ Phase 2 – End-to-End Workflow Implementation Design

Specification

Purpose: Provide a developer-ready implementation blueprint for the GuardianIQ governed workflow scheduling, AI agent execution and monitoring capabilities. This document is intended to be the primary implementation reference for frontend, backend, database and QA teams.

1. Scope and Objectives

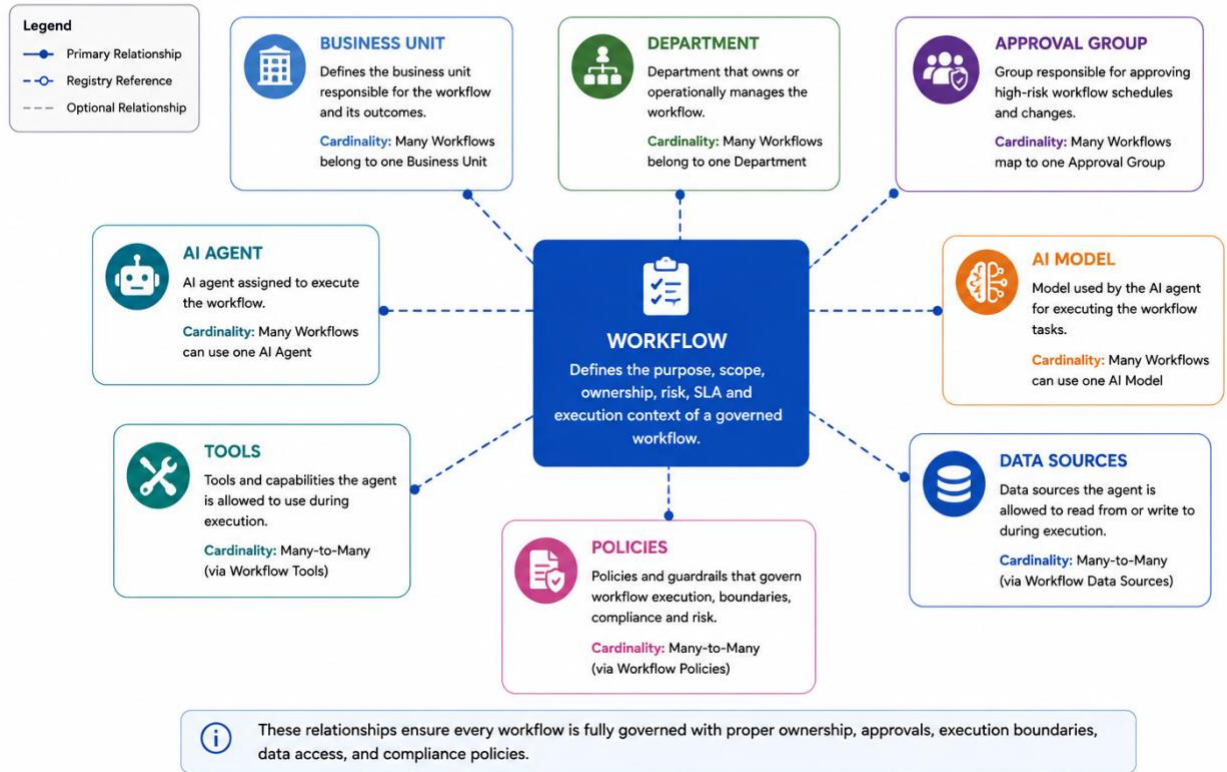
- Implement governed workflow scheduling and execution capabilities on top of the completed Phase 1 Registry foundation.
- Support workflow definitions, schedules, workflow runs, AI agent assignments, approvals, execution boundaries, audit trails and notifications.
- Provide end-to-end traceability from workflow creation to execution outcomes and audit.

2. Target Capabilities

- Workflow Registry
- Schedule Management
- Scheduler Engine
- Workflow Run Engine
- AI Agent Boundary Enforcement
- Audit & Notification Services
- Run History & Monitoring
- Role-based UI and APIs

Workflow Registry Relationship Diagram

Core relationships and dependencies managed within the Workflow Registry

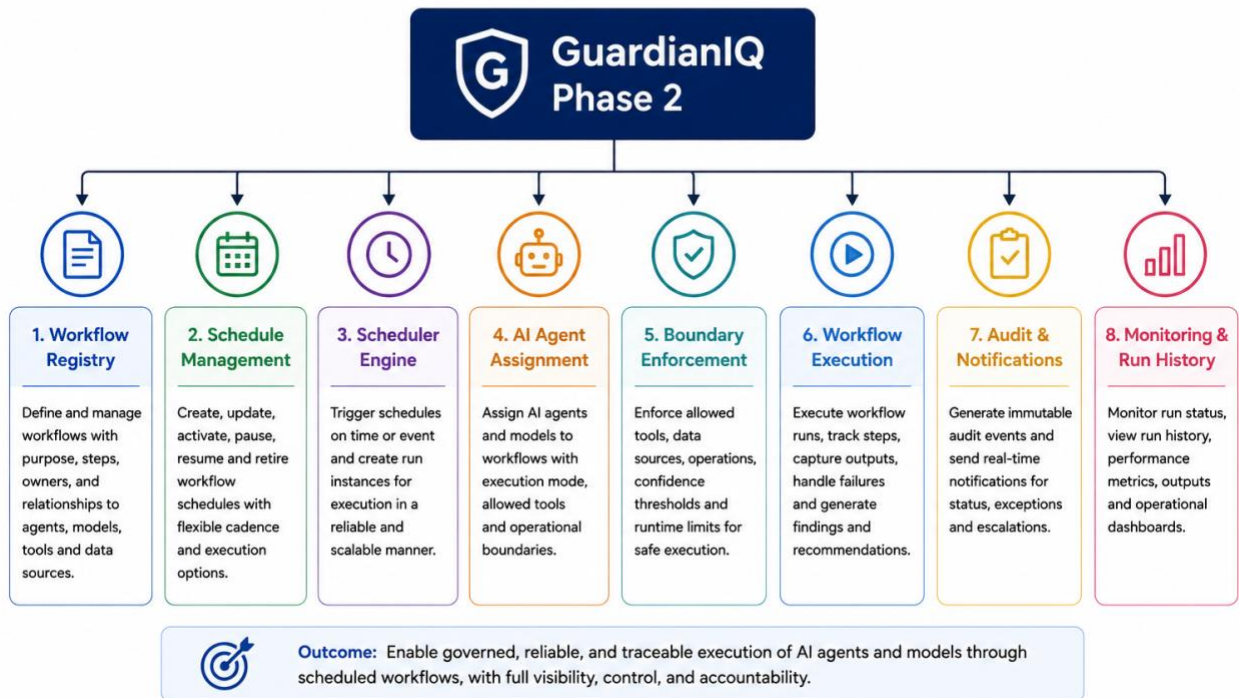


3. End-to-End Workflow Lifecycle

- Create Workflow → Assign AI Agent → Configure Boundaries → Configure Schedule → Configure Governance Controls → Approval → Activate Schedule → Scheduler Triggers Run → Boundary Validation → Agent Execution → Findings & Recommendations → Persist Results → Audit & Notifications.

GuardianIQ Phase 2 – Capability Map

Core capabilities delivered in Phase 2



4. Backend Services

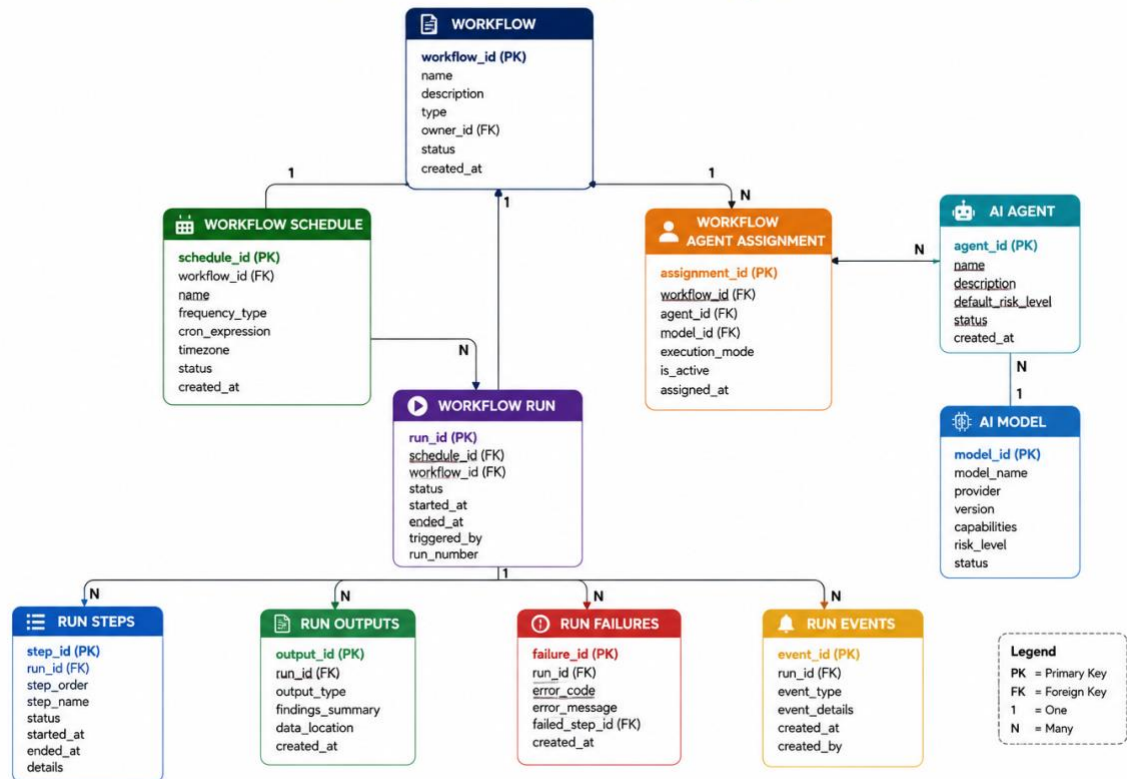
- Workflow Service
- Schedule Service
- Scheduler Service
- Execution Service
- Boundary Service
- Authorization Service (RBAC + ABAC)
- Audit Service
- Notification Service
- Monitoring Service

5. Database Components

- workflow_schedules
- workflow_runs
- workflow_run_steps
- workflow_agent_assignments
- workflow_execution_logs
- workflow_outputs
- workflow_failures
- workflow_notifications

- workflow_events

Workflow Database ER Diagram



6. API Design

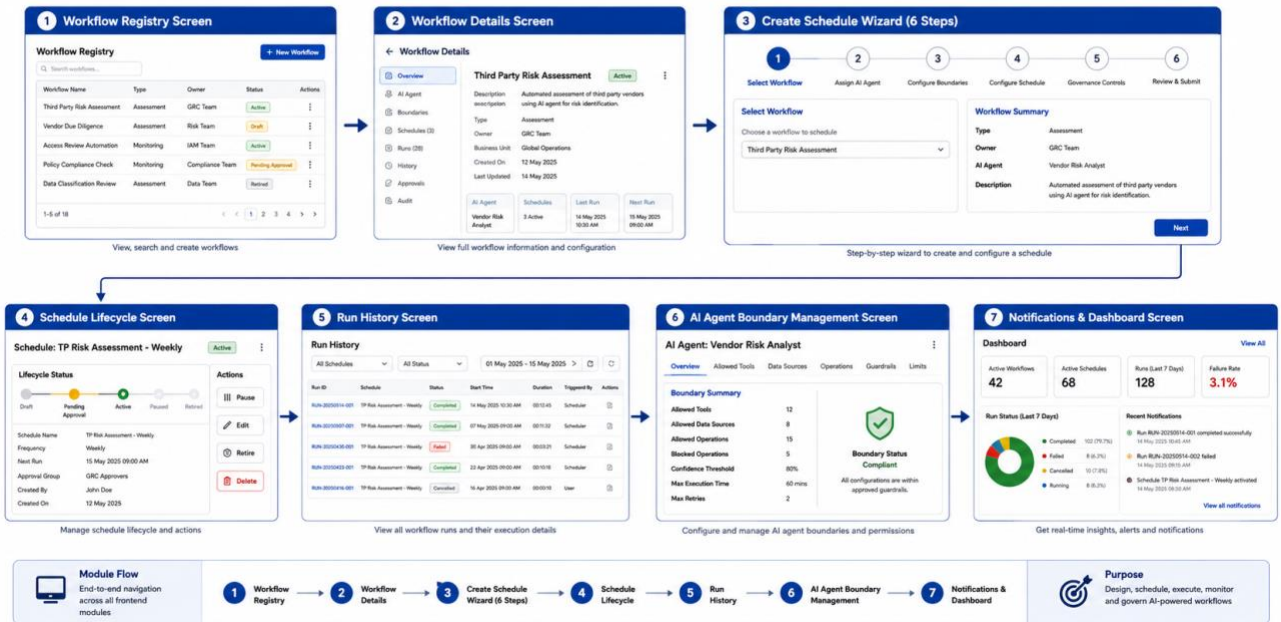
- Workflow APIs: Create, Update, Delete, Search, Versioning
- Schedule APIs: Create, Activate, Pause, Resume, Retire
- Execution APIs: Trigger Run, Retry, Cancel
- Run APIs: View History, Outputs, Logs
- Monitoring APIs: Metrics and Dashboard feeds

7. Frontend Modules

- Workflow Registry Screen
- Workflow Details Screen
- Create Schedule Wizard (6 Steps)
- Schedule Lifecycle Screen
- Run History Screen
- AI Agent Boundary Management Screen
- Notifications & Dashboard Screen

GuardianIQ Phase 2 – Frontend Modules (Integrated View)

End-to-end user journey across all key frontend modules



8. Detailed Schedule Wizard Steps

- Step 1: Select Workflow
- Step 2: Assign AI Agent and Model
- Step 3: Configure Allowed Tools and Data Sources
- Step 4: Configure Schedule and Retry Policies
- Step 5: Configure Governance Controls, Owners and Approval Groups
- Step 6: Review and Submit

9. AI Agent Boundary Controls

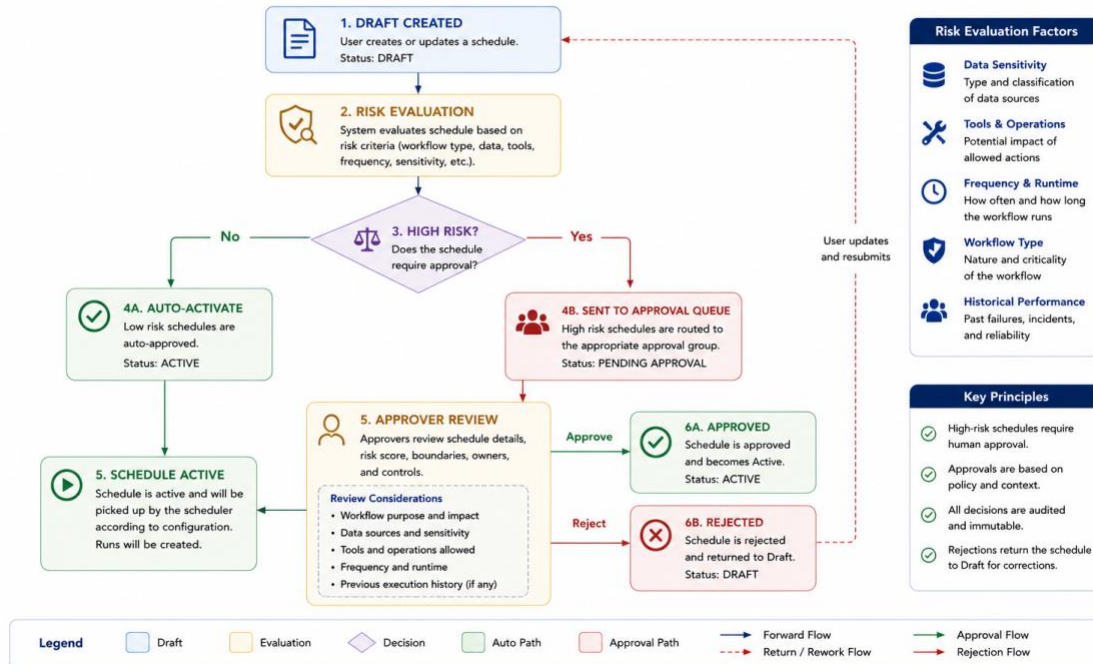
- Allowed Tools
- Allowed Data Sources
- Allowed Operations
- Blocked Operations
- Confidence Threshold
- Maximum Runtime
- Retry Limits
- Human Approval Rules

10. State Machines

- Schedule States: Draft → Pending Approval → Active → Paused → Retired
- Run States: Queued → Running → Completed / Failed / Cancelled / Skipped

12. Workflow Approval Process

Governance-controlled approval flow for workflow schedules



11. Security and Governance

- RBAC and ABAC authorization
- Immutable audit events
- Multi-tenant readiness
- Sensitive data restrictions
- Delegation and approval support

12. Audit Events

- WORKFLOW_CREATED
- SCHEDULE_CREATED
- SCHEDULE_ACTIVATED
- RUN_STARTED
- BOUNDARY_VALIDATED
- AGENT_EXECUTED
- FINDINGS_GENERATED
- RUN_COMPLETED
- RUN_FAILED

13. QA Scenarios

- Positive flow tests
- Boundary violation tests
- Approval workflow tests

- Failure and retry tests
- Performance tests
- Security tests

14. Definition of Done

- All APIs functional
- All screens integrated
- Audit events generated
- Scheduler and runs working end-to-end
- QA sign-off complete
- UAT approved

15. Recommended Implementation Sequence

1. Finalize database schema and migration scripts.
2. Implement repositories and ORM models.
3. Develop backend services and APIs.
4. Implement scheduler engine and workflow execution engine.
5. Implement audit and notification services.
6. Develop frontend screens and wizard.
7. Integrate APIs and frontend.
8. Execute QA and UAT.
9. Deploy and hand over.